

Exploring Potential Contributions of Social Learning to Adaptive Learning Systems

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ABSTRACT

Adaptive learning systems aim to emulate how skilled educators seek to provide every student the best possible learning experience. We investigate how such systems might be enriched by activities and indicators of social learning - an aspect of learning that focuses on the influences of learners' social context and interactions. In this paper we describe a pilot study aimed at exploring the potential for including social learning in an adaptive system. Our analysis of the social learning scale demonstrates its validity and usefulness for our ongoing work. Our qualitative analysis of students' learning demonstrates how social learning vary among students. We discuss how the rating scale results and observations of social learning can be integrated within a student model needed to drive an adaptive system. More generally, our work illustrates how theories of learning can contribute to the design of adaptive learning systems.

CCS CONCEPTS

• **Human-centered computing** → *Computer supported cooperative work*; • **Applied computing** → **Interactive learning environments**.

KEYWORDS

adaptive learning, social constructivism, education, peer learning

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1 INTRODUCTION

For the past 25 years, research on adaptive learning systems (ALS) has sought to model learners' knowledge in a way that enables adjustments to the learning activities available or encouraged, such that different learners follow personalized learning pathways [20]. The general approach has been to build a domain model for the knowledge or skill being acquired (e.g., algebra), assess the novice learner against that model to create a student model, then build the individual model to reflect the learning that takes place as the

student completes specific learning activities, using evidence from relevant assessments (e.g., successful problem completion versus errors) to adjust what has been learned (or not learned) at each point [29].

The ALS approach has yielded some recognized successes¹. During adaptation, the system dynamically reorganizes learning resources to meet specific learning objectives based on an individual learner's profile or learning portfolio [9] [35]. However, while general progress has been made in developing and applying ALS technologies, the example systems reflect a rather narrow view of learning, with a central focus on the *content* of the knowledge domain being taught (e.g., algebra, biology [27]). Attention has focused on the technical issues of what data to collect, how to model it, and what prediction methods should be used to decide what would be the best next step for the individual [22].

We argue that ALS researchers should adopt the richer view of learning that has emerged in research on education, where there is an emphasis on cognitive processes that evoke and promote learning in the real world [6]. In particular, modern theoretical accounts of human learning emphasize the important contributions of the learner's social context, in combination with reflections about his or her knowledge state and learning strategies (metacognition [15]). Our research seeks to address this gap in ALS frameworks by exploring how one might assess and integrate learners' social learning and metacognition as part of an adaptive process.

The current paper reports a pilot study used to develop and validate a quantitative measure of social learning that could be integrated within an adaptive system. The pilot study also designed peer-learning components of classroom assignments that should both elicit and benefit from social learning processes. We discuss how we might use a combination of the rating scale data and scripted prompts that encourage peer learning as input into an adaptive learning system.

2 BACKGROUND

2.1 Social Learning in the Classroom

Social constructivism is a specialization of constructivism that emphasizes a learners' interactions with the social context in which they are embedded [36]. Social learning theory emphasizes the impacts of other learners ("peers") on an individual [5], where learning often occurs through direct or incidental observation of peers' behaviors and language [17] [28]. Learners may model the behavior enacted by more advanced peers; or, they may recognize a peer's understanding as an alternate view of a concept they are

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¹<https://www.emergenresearch.com/blog/top-10-adaptive-learning-solution-providers-in-the-world>

building. In either case, the learner’s understanding is enriched. Comparing one’s progress to peers evokes self-reflection and adjustment, which are metacognitive processes. In addition, responding to and offering peer corrections and suggestions may help learners to discover problems or weaknesses in their own work [25] [21].

Educators have begun to apply the theories of social learning to their classroom teaching, for example using “think aloud” as a technique for modeling internal cognition in a way that can be shared with others and serve as a resource for reflection [7]. A related application is the research on peer learning, wherein students interact with peers to attain educational goals [19] [24]. Frayne and Laytham showed that learners in a professional development class benefited greatly from peer instruction and learned valuable skills in how to seek and give support [17].

Peer-based learning activities are common in the university setting, as instructors often incorporate team projects and other forms of team activities within their courses [23]. Although a team setting may not include an “expert” as a peer, it is quite likely to include students who vary in expertise on a given topic. This leads to opportunities for peer observation, comparison and social learning. In the current work, we are leveraging the popularity of team-based learning activities in university classes, to set up a situation conducive to our exploration of social learning within an adaptive system.

2.2 Assessing Social Learning

Most research seeking to evaluate socially-mediated learning has been qualitative. Educators have noted that behaviorist learning and teaching interactions could produce a culture of learners’ dependence on teachers; the students do as they are told and exhibit good surface understanding, but little sense of purpose [37]. In contrast, social constructivist approaches acknowledge the need for student-student interaction. Not surprisingly, the language-based interaction among peers in a classroom setting is difficult to track, and little if any work has been directed at developing more convenient and repeatable measures such as the rating scales that have been developed to assess metacognition [16]. Our project takes a step in that direction by designing a self-report survey instrument.

One successful example of quantitative assessment in the realm of social learning is found in Albert Bandura’s work on self-efficacy [2]. Self-efficacy refers to an individual’s beliefs about their own capacity for succeeding with a particular task or activity [3]. Importantly, the construct is distinct from a general sense of achievement, tied instead to expectations for a given domain or topic. Like other constructs posited by Bandura, self-efficacy is socially mediated, for example through modeling (if she can do it, I can do it) and feedback (she says I am doing it wrong). Bandura provides considerable guidance for and examples of how to assess self-efficacy in a variety of domains (e.g., math, cooking, weight loss) using rating scales [4]. One result is that self-efficacy scales have been developed and validated for many different domains of learning and are often used as surrogates for measures of achievement [33].

2.3 Adaptive Learning Systems

Adaptive learning aims to provide suitable and multiple learning materials and paths based on the learner’s characteristics (e.g.,

learner behavior, learning preferences, learning styles, prior knowledge) [8] [14] [26]. In university education, many instructors have come to rely on learning management systems (LMS; e.g., Canvas²), with the result that many materials, activities, and learning interactions take place online. This situation is useful for ALS: as learners interact with different online learning systems, they leave a considerable amount of data logged by the system, which contains valuable information about their learning process.

A core requirement for ALS is to collect, organize, analyze and make sense of learner activities (e.g., using interaction logs, records of content access or use, communication with the instructor or peers). The hope is to grasp the learners’ behavioral or learning patterns that reflect whether, what, and in what ways the learner is responding to instruction [1, 27]. If learning objectives can be decomposed and synchronized with the learner models, a learner-specific trajectory can be planned and implemented. If an ALS is made available for an individual (or related) courses, the instructors can offer personalized instruction that meets the needs of all learners [10].

One open issue for design of ALS is to characterize (and model) the pre-existing differences among learners that might impact their learning success. For example, modern theories of education highlight ideas such as situated cognition, which posits that learning takes place through cognitive and social processes that are bound to social, cultural, and physical contexts [6]. Under this lens, learners with differing socio-cultural backgrounds are likely to approach their learning goals differently [18], suggesting that the development of ALS learner profiles should consider more than a straightforward assessment of content knowledge (e.g., algebra, Spanish; [11]). Our research goal is to move in this direction, by considering the social learning tendencies that may affect individuals’ progress through a learning activity. The research is exploratory in nature, with aims focused on assessment of social learning tendencies, as well as evoking and social learning in a way that can provide input to an adaptive learning process.

2.4 Research Approach

Empirical studies have demonstrated the value of adaptive systems that track and analyze learner data collected from online learning systems. The process to ‘test’ the progress of learning objectives is relatively straightforward as it is tied to the content being delivered. Individual grades for tests or other assignments, or perhaps key errors made in completing assignments can all serve as indicators of learning. In contrast, it is much less obvious to measure how (and how much) learners vary in their social learning strategies, or how they use such processes for their learning [13]. To explore these issues, we needed to design an educational setting that can evoke and potentially benefit from social learning, so that they can be observed and documented in a way that suggests individualized adjustments to the learning context and activities (i.e., to support adaptive learning).

To be useful, the educational setting should have several characteristics: it should offer learning goals and activities in which social learning may contribute to success; it should include instruments or indicators that capture learners’ initial and ongoing practices of

²<https://www.instructure.com>

social learning; it should document the corresponding variability within the population of learners. In the following sections we describe the setting we assembled as part of an undergraduate course on information visualization. We follow this with a description of the data that we collected, both with respect to the rating scales used to assess social learning tendencies, and the qualitative data gathered during the learning activities. As related constructs, we also report results from an established scale for metacognition [31] and a custom scale created to measure self-efficacy for information visualization. We conclude with a discussion of how the pilot study has prepared us for design work aimed at integrating social learning into ALS.

3 METHODS

Our eventual goal is to design an ALS that can be used as part of university courses, so we first identified a suitable course. We worked with the instructor to design learning activities that would contribute to the course’s objectives and allow us to investigate the social learning processes of interest. We developed a self-report survey for assessing students’ beliefs about their social learning tendencies. The following sections describe these steps in more detail.

3.1 Social Learning, Metacognition and Self-efficacy Scales

Creating a social learning scale (SLS) was a primary goal for this project. No published instruments were available to use as a starting point, so we created a new set of items. We reasoned from theories of social constructivism [36] and social learning [2] to brainstorm 18 items. The items focused on students’ skills and successes when working with peers (e.g., “Working in a group allows me to focus on parts of the task I do best, and learn from others on the parts they do well.” and “I feel more motivated to learn when I am working in a group.”). Prior to the classroom study, we carried out an informal comprehension test of the resulting items and found no issues.

For metacognition (MCA), we reused the Metacognition Awareness scale [31], which has 45 items. For self-efficacy (SE), we modeled on the many examples of domain-specific self-efficacy [4] to create 10 items probing capacity for evaluating and designing information visualizations.

3.2 The Course Setting

The pilot study took place within an undergraduate course that introduced information visualization. A single instructor taught two sections of an in-person class that met three times a week. We invited participation by students across the two sections. Participation was voluntary and included signing informed consent. The participants did not have any special prior experiences with peer learning. They were told that participation included one survey (a combination of SLS, metacognition and self-efficacy) as well as participation in several activities that had been designed to explore students’ learning processes. They were also told that we would gather classroom observations during the activities. We emphasized that we were seeking information about how to build systems that would better support the individual needs of students like them. 17 students across the two sections agreed to participate.

Table 1: Means and Variability for MCA, SE and SLS

	N	Mean	Min	Max	Std Dev
MCA	97	3.67	2.47	4.67	0.42
SE	96	3.68	2.00	5.00	0.64
SLS	97	3.76	1.89	5.00	0.69

Over four weeks, the students who agreed to participate completed the surveys and participated in the learning activities summarized below. We use their responses from the survey, observations taken in classes, and at times review of their assignments in making inferences about the presence and role of social learning and related metacognitive processes. During a subsequent semester, we collected additional responses to the survey items, to improve the power of statistical analyses conducted on the scales. These data were collected a new population of students enrolled in the same information visualization course and produced an additional 81 data records. No additional observational data were gathered at this time.

3.3 Learning Activities

We collaborated with the instructor to design two activities that might evoke social learning. Both were examples of peer-based learning, leveraging the general approach of peer review. We focused on peer-review because past research indicates that these activities help students reflect and compare their performances with their peers [30]. Studies have also indicated that peer review leads to improved final learning outcomes, even in the online environment [34][12].

The goal *Activity 1* was to provoke self-reflection in reaction to the evaluation of a peer’s assignment. In this case, the reviewed assignments were graphs created to visualize an instructor-provided dataset. Students were asked to assess and write feedback comments for graphs designed by peers. In contrast, the goal of *Activity 2* was to promote peer interaction in the context of a group assignment. Students were assigned to groups and asked to discuss the feedback and comments created in the first assignment; they were asked specifically to comment on whether and how the visualizations made use of the design guidelines provided by the instructor. Class observations were recorded to document the interactions in this group activity.

4 RESULTS

The pilot study had two primary goals: to evaluate the usefulness of the SLS rating scale in a classroom setting; and to observe the effectiveness of two peer-based learning activities in evoking social learning experiences. We first report our quantitative findings and follow that with a qualitative analysis of the classroom observations.

4.1 Students’ Self Perceptions

Three categories of rating scale data (metacognition (MCA), self-efficacy for information visualization (SE), and social learning tendencies (SLS)) were gathered from multiple sections of an information visualization courses; all valid data records were combined to create a complete dataset of 97 records. The 45-item well-tested

Table 2: Items and Exploratory Factor Loadings for Social Learning Scale

	Factor 1	Factor 2	Factor 3
One benefit of group projects is that they help me reflect on my learning	0.798	0.198	0.013
I feel more motivated to learn when I am working in a group	0.760	0.265	0.154
I am effective when learning in a group	0.723	0.393	0.222
In a group learning setting, I find it helpful to know how much time other people spend with the coursework	0.667	-0.028	0.026
I find it useful to observe other people's learning processes when I am working with them	0.662	0.367	0.310
I enjoy doing my coursework more when I am working in a group	0.646	0.422	0.052
When I work in a group, I find the experience rewarding	0.581	0.640	0.091
Working in a group allows me to focus on parts of the task I do best, and learn from others on the parts they do well	0.550	0.521	0.087
I benefit from the interactions during shared problem solving in a group	0.500	0.482	0.445
When I work in a group I find it easy to fit the group work into my schedule	0.420	0.507	-0.201
When I work in a group, it is overall better than working by myself.	0.398	0.741	-0.019
When I work in a group other people's learning helps in the evolution of my personal views	0.365	0.598	0.089
When working in a group project. I frequently check on my partner's progress	0.300	-0.163	0.659
When I work in a group I am able to focus on the task at hand for a longer period	0.187	0.790	0.016
I believe the work I submit as a part of a group is of higher quality than what I would have submitted by myself.	0.178	0.789	-0.090
When I am a part of the group, I feel personally involved.	0.154	0.092	0.867
When I work in a group I try to keep the conflicts to a minimum	-0.060	0.117	0.690
One benefit of group projects is that they help me reflect on my learning	-0.131	0.691	0.439

MCA scale [32] demonstrated strong internal reliability ($n=87$; Cronbach $\alpha = 0.92$). The 10-item SE scale and 18-item SLS scale also demonstrated strong internal reliability ($n=95$, Cronbach $\alpha = 0.90$ for SE; $n=97$, Cronbach $\alpha = 0.95$ for SLS). These findings indicate that the SLS scale should be a useful measure for indexing students' differential tendencies toward social learning.

Given the internal reliability of all three scales, we created constructs to document the distribution for each measure. The corresponding means and standard deviations for each measure are shown in Table 1. Two important observations are that all three measures capture the variability among students (i.e., as evinced in the range), and that the standard deviations are modest. Histograms of the frequencies confirmed relatively normal distributions with a peak in each case around the mean. Exploratory correlation analysis also revealed that the three measures are highly correlated (ranging from 0.499 for MCA and SE to 0.288 for SLS and SE, all significant at $p < .01$).

Because a primary research aim was to explore the assessment of social learning, we conducted additional exploratory analyses on the SLS scale, using factor analysis to uncover any underlying internal structure. This analysis yielded a three-factor solution accounting for 61.5% of the variance; the factor loadings for each scale item are listed in Table 2, with items ordered by size of the loading on the first factor. The Varimax rotation was used to aid interpretation; we focused attention on items that loaded strongly on one factor with little impact on other two. Our tentative interpretation is that Factor 1 may capture social learning tendencies related to *learning motivation* (e.g., "I feel more motivated to learn when I am working in a group"). Factor 2 may instead be capturing variations in *learning outcomes* (e.g., "I believe the work I submit as a part of

Table 3: Students Featured in Social Learning Behavior Analysis

	Kyle	Tim	Kelly	Lisa	Ruth
SLS	4.06	3.56	3.44	3.22	2.28

a group is of higher quality than what I would have submitted by myself"). The final factor has fewer items for interpretation but may be related to *social interaction* (e.g., "When I am a part of the group, I feel personally involved"). We stress that these interpretations are highly speculative, but for researchers interested in a more articulated assessment of social learning tendencies, they suggest a path for future study.

4.2 Observations of Social Learning

In this section, we shift focus to the social learning activities that we observed, with discussion of how our observations might help us to track or evoke such behavior within an ALS. To support this research aim, we consider the observations of five students who have a wide range of initial SLS scores Table 3 lists a pseudonym for each student, along with his or her SLS score. We chose these students because their SLS scores ranged from relatively low to relatively high, and we had notes about some of their classroom activities.

4.2.1 Peer Review. In the first activity, the students carried out and submitted a review of a peer's work; they were assigned to carry out a critical analysis of graphs that had been submitted. Kyle's critical review indicated great thought, engagement, and detailed observations. This indicates a good grasp of content and

attention to detail for the peer he reviewed. When asked, he also indicated that doing the peer review helped him to reflect on his visualization project. These observations are consistent with his self-reported high SLS scores; he benefits from peer interactions and can recognize this positive impact.

Other observations focused more on the benefits of peer input. For example, Kelly (moderate SLS) received feedback praising her project with adjectives such as “exceptional” and “perfect.” These positive comments were paired with a detailed analysis of what worked well. Kelly felt encouraged by this peer review, and an ALS system could update its model of her social learning accordingly. For example the system might ask her to respond to a prompt her after receiving the review, e.g. “How did you feel about the feedback you received?”

A contrasting case can be seen in Lisa’s experience (moderately low SLS). She did not receive feedback that was detailed, and as a result, she did not gain many benefits. However, she was still able to consider whether and how the feedback might influence her thinking in the future. In this case, Lisa noted that because it was limited in detail, it was not helpful as a model for her future work. To provide useful feedback to a peer, the reviewer must understand the relevant concepts enough to explain how they apply. This suggests that more successful students (who are doing well with the material) should be assigned as peer reviewers for those who have been less successful.

More generally, we observed that students’ feedback to others was guided by what they believed to be necessary for improvement, which reflects their own work and understanding. We can see that learners who scored highly on SLS (e.g., Kyle, Lisa) were analytical of others’ work and tended to reflect on how their consideration of other assignments affected their own understanding.

4.2.2 Peer Interaction. In the second activity, the students worked in teams during class to critically analyze two graphs provided by the instructor. The instructor chose examples such that one visualization followed the graphic design principles he had just presented, while the other did not. Students generally found this activity engaging, with much discussion and suggestions distributed across the groups.

One of the groups included both Kyle and Kelly. That group adopted a relatively systematic approach to peer interaction: they assessed the two example projects individually, then met together to discuss and develop a group report. The group went through a detailed account of strengths and weaknesses using the guidelines they had been given. The contributions of Kyle and Kelly - who both had moderately strong SLS scores - helped the group stay on track. In Lisa’s group, students also began by reviewing the examples individually. However, in contrast to Kyle and Kelly’s team, this group reached out to the instructor for an extended conversation, perhaps because they had less expertise in synthesizing peer input.

We noted that a few groups viewed the assignment as an individual task. They did not connect ideas from their peers to their own work; rather they created lists of ideas individually and appended them in a simple fashion. These dynamics suggest a possible role for an ALS that notes the absence of peer interaction (assuming that interactions were taking place online), and prompts students to discuss ideas with peers while writing the shared report.

5 DISCUSSION

We have presented findings related to two main research goals in the current work: 1) We created, tested, and validated an SLS instrument for assessing social learning tendencies across students; and 2) We discussed how variations in SLS might relate to the types of social learning behaviors students emit when engaged in peer-based review and interaction.

The new SLS construct can be used to assess - and potentially track over time - a student’s confidence and successes with social learning activities. For example, while the 18-item scale might be used to characterize students at the start of a course, an abbreviated version might allow an ALS to track changes in social learning capacities (and guide associated interventions) over the course of multiple learning activities.

We documented normal variation in student scores for social learning, which can be used to drive decision-making in an ALS. Variations in SLS point to an opportunity for adaptive support that might improve the social learning experience and outcomes for students. For example, our observations of Kyle suggested that doing a critical review of a fellow student’s work helped him to reflect on his own. A student with a relatively low SLS score might be prompted by an ALS to carry out similar reflections.

The class observations also have implications regarding the creation, evolution and use of student models in an ALS. For example, we argued that ALS-driven pairing of students for peer review could use a combination of current class performance and SLS score, with students who are doing well (and who have high SLS) providing critical reviews for their peers. The process would help both - the stronger student would practice communication of key concepts (e.g., “I think I should explain it like this”) while the weaker student would benefit from having a successful role model (e.g., “Now I see how to understand that issue”).

At a more granular level, an ALS might use a combination of SLS indices (perhaps abbreviated for in-class assessment) and capture of social behaviors (e.g., frequency of text messages to peers) to guide adaptation of the student models and subsequent interventions. For example, a student who is operating at a relatively “low” level of social learning (e.g., Ruth) might be prompted reach out to a peer or to reflect on feedback that was recently received.

6 LIMITATIONS AND CONCLUSIONS

The work presented here reflects an initial step toward the integration of social learning in adaptive systems. However, our dataset is relatively small (especially the behavioral observations), so our findings must be interpreted with caution. The SLS scale has strong validity but may benefit from additional exploration of distinct subscales. Meanwhile, we are continuing in this line of research. Our current activities are aimed more directly at ALS design; we are using Wizard of Oz methods to track, intervene and promote social learning behaviors at an individual level. The preliminary work reported here has provided a critical foundation for this ongoing research.

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